Vector Model Of Atom: Different Quantum numbers and their symbols:

1. Principal Quantum no: (n): This governs the total energy E_n

$$E_n \propto \frac{-1}{n^2} \text{ where } n = 1, 2, 3 \ldots\ldots$$

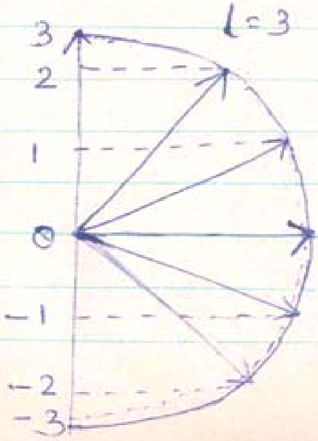
2. Orbital Quantum Number (l): This has values $l = 0, 1, 2 \ldots (n-1)$ for each n and governs the orbital angular momentum of magnitude

$$|\vec{P_l}| = |\vec{l}| = l^* \hbar = \sqrt{l(l+1)}\,\hbar,$$ l is always +ve or zero.

3. Spin Quantum No: (s): This can only have one value $s = \frac{1}{2}$ (never put $s = -\frac{1}{2}$) and relates to the intrisic property of electron which we call its spin angular momentum. Its value

$$|\vec{P_s}| = |\vec{s}| = s^* \hbar$$

where $s^* = \sqrt{s(s+1)}$ $\therefore$ $P_s = \sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\,\hbar = 0.866\hbar$ for one electron.



$|\vec{P_l}|_z = l_z = m_l \hbar$ where m_l is an integer; m_l has $(2l+1)$ values from $-l, -(l-1), 0 \ldots (l-1), l$ in unit steps.

(4) The ~~total~~ orbital angular Quantum No (ms
This relates to the magnetic fild of
Z component of the orbital angular
momentum vector $|\vec{l}|$ through

$$\left|\vec{P}_l\right|_z = l_z = m_l\hbar \text{ where } m_l \text{ is an intege}$$

The possible values of m_l are

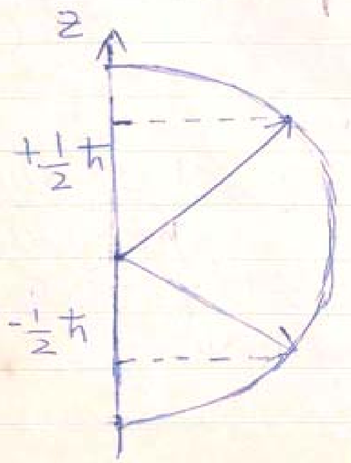
$$m_l = -l, -(l-1) \cdots 0, (l-1), +l \cdots$$

ie $(2l+1)$ possible values of m_l for each value of l.

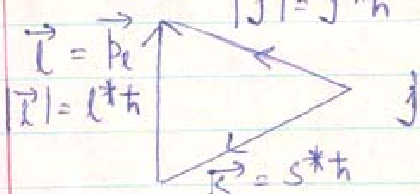
(5). <u>The Spin magnetic Quantum no: (m_s)</u>:
This depends on the magnitude of the z component of the spin angular momentum $\vec{P}_s = \vec{S}$ through $S_z = m_s\hbar$ where m_s can be $+\frac{1}{2}$ or $-\frac{1}{2}$.

$$\vec{P} = s^*\hbar = \sqrt{s(s+1)}\ \hbar = 0.866\hbar$$

$$(\vec{P}_s)_z = S_z = m_s\hbar$$



(6) Total angular momentum vector : For a Single electron vector $\vec{l}$ i.e $\vec{P_l}$ and $\vec{s}$ i.e ($\vec{P_s}$) Couple vectorially in a non- Strong field to give vector $\vec{J}$ i.e ($\vec{P_j}$) where $\vec{J} = \vec{l} + \vec{s}$
i.e $\vec{P_j} = \vec{P_l} + \vec{P_s}$
The numerical value of $\vec{J}$ i.e ($\vec{P_j}$) Can be obtained from the triangle of vectors of numerical value l^*, s^* & j^*



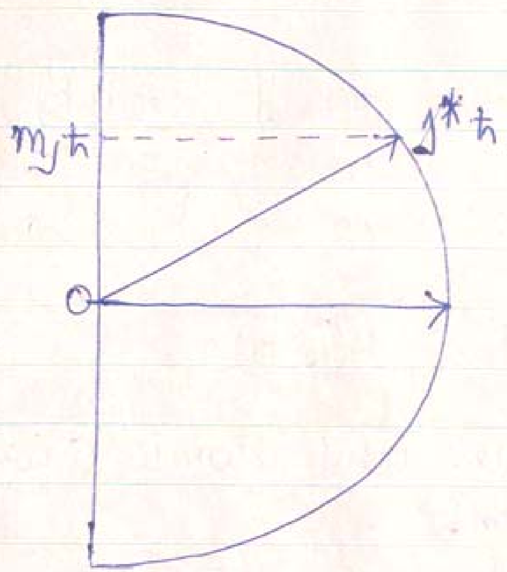
$j^* = \sqrt{j(j+1)}$

j = The total atomic Quantum number.
$j = l+s$ & $(l-s)$

It is to be noted that the angular momentum vectors $\vec{l}$ & $\vec{s}$ are never parallel nor anti parallel to each other. $\vec{J}$ gives the total angular momentum.

(7) The total Quantum number of the electron $\vec{J}$ given by $j = l + \frac{1}{2}$ and $l - \frac{1}{2}$ i.e two values for each l in zero or weak field. The Quantum number j relates to the magnitude of the total angular momentum through $|\vec{j}| = j^*\hbar = \sqrt{j(j+1)}\,\hbar$ where j is positive and always $\frac{1}{2}$ integral for a Single electron. Each l level degenerates into two j levels.

(8). The total magnetic Quantum number m_j relates to the component of the total angular momentum vector through $J_z = m_j \hbar$ where m_j can have $(2j+1)$ values from $(-j$ to $+j)$ in unit steps.



(9) For a given n, value, we have values for l from 0, 1, 2 - - - $(n-1)$ and & the sub orbital are named as s, p, d, f - - - - - for $l = 0, 1, 2, 3$ - - - respectively.

(10) Selection-rule: when an electron jumps from one level to another. it must be in accordance with the following restrictions

For Q.N n ; $\Delta n = (0), 1, 2, 3$
For Q.N l ; $\Delta l = \pm 1$.
For Q.N s : $\Delta m_s = 0$
For Q.N j : $\Delta j = 0, \pm 1$ but $j_1 \to 0$ to $j_2 = 0$ is forbidden.

(11) The above selections rules are found to be appropiate where there is a strong spin-orbit coupling, $j = l \pm \frac{1}{2}$ i.e when the external magnetic field is very weak or zero. But if the spin-field (external magnetic field and the orbit field interactions are strong, the above rules are not quite strictly followed).

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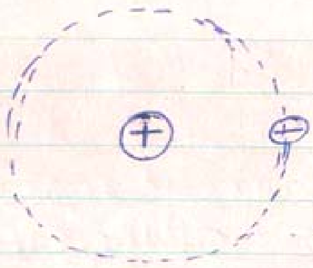
The magnetic moment of an electron in an atom:-



Let r = radius of the circular path.

ω = The angular velocity of rotation of the electron in the orbit.

$T = \frac{2\pi}{\omega}$ = The time taken to make one rotation in the orbit.

e = charge of electron.

m = mass of electron.

The rotating charge, in a closed circular path would give rise to a current $i = \frac{Q}{t}$

$$i = \frac{e}{T} \quad \text{—— (1)}$$

This current loop is equivalent to a magnetic shell, having magnetic moment $\mu_l = iA$ —— (2)

Equation (2) gives the magnetic moment due to the orbital motion of the electron.

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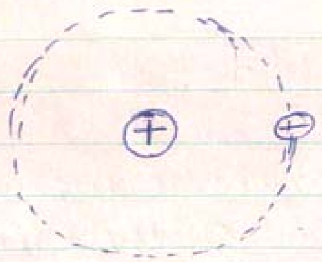

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or $\mu_l = \frac{e}{T} \times \pi r^2$ —— (3) | $A = \pi r^2$.

The electron rotating in the circular path possesses an angular momentum.

$$I\omega = mr^2 . \frac{2\pi}{T} = \vec{p}_l = \vec{l^*}\hbar.$$

$$\text{or} \quad \frac{\pi r^2}{T} = \frac{\vec{p}_l}{2m} = \frac{\vec{l^*}\hbar}{2m} \quad \text{—— (4)}$$

Putting eq^n (4) in (3) :-

$$\mu_l = \frac{e}{2m} . p_l = \frac{e}{2m} l^* \hbar \quad \text{—— (5)}$$

Equation (5) gives the magnetic moment due to the orbital motion of the electron.

$$\mu_l = \frac{e\hbar}{2m} \vec{l^*} \text{ Joule / Tesla.}$$

| $W = MB(1-\cos\theta)$ |
| $M = \frac{\text{Joule}}{\text{Tesla}}$ |

but $\frac{e\hbar}{2m} = \mu_B$ = Const. = Bohr-Magneton

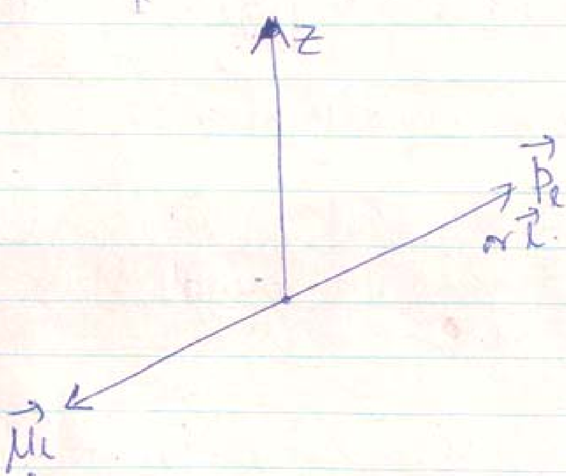
$$\mu_B = \frac{1.6 \times 10^{-19} \times 1.05 \times 10^{-34} \text{ J-sec}}{2 \times 9.1 \times 10^{-31}}$$

$$= 9.27 \times 10^{-24} \text{ Joule/Tesla.}$$

$$\therefore \boxed{\vec{\mu}_l = \mu_B \vec{l^*}} \quad \text{—— (6)}$$

We Know that $\vec{l^*}$ (i.e $\vec{p_l}$) is along that dir if we curl the fingers of right hand in the direction of rotation of the electron the thumb gives the direction of $\vec{p_l}$ & $\vec{l^*}$.

Since charge of electron is −ve, it gives a current opposite to the direction of rotation of the electron. ∴ If we curl the fingers of right hand in the direction of current the thumb gives the $\vec{\mu_l}$. Hence $\vec{\mu_l}$ & $\vec{p_l}$ are oppositely directed.



Equation (6) is more correctly written as

$$\vec{\mu_l} = -\mu_B \vec{l^*} \quad \text{--- (7)}$$

$$\vec{\mu_l} = -\vec{l^*} \text{ in units of Bohr-Magnetor} \quad \text{--- (8)}$$

$$\vec{p_l} = l^* \hbar = l^* \text{ in unit of } \hbar \quad \text{--- (9)}$$

Ratio of the magnetic moment in units of Bohr magneton and the orbital angular momentum in units of ħ is known as Gyromagnetic ratio.

Gyromagnetic ratio $g_l = \dfrac{\text{Magnetic moment in units Bohr magneton}}{\text{Orbital angular momentum units of } \hbar}$

$$g_l = \frac{l^*}{l^*} = 1$$

Although in this simple case of one electron system $g = 1$ but there are cases where $g_l > 1$ $\therefore$ $\boxed{g_l \geq 1}$ Hence

Equation ⑦ can be written as:

$$\boxed{\mu_l = -g_l \mu_B l^*} \text{ --- } (10)$$

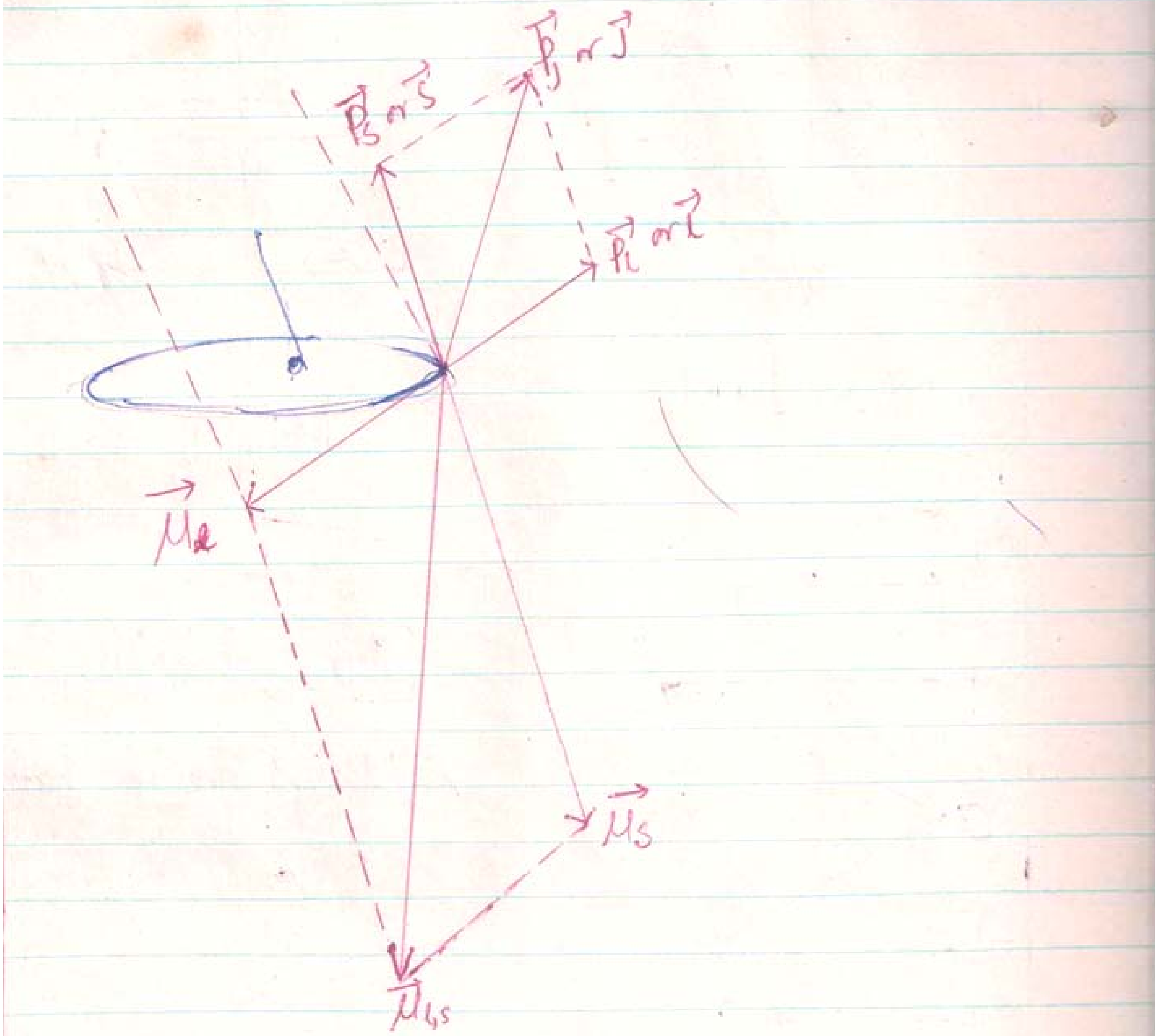
Spin angular momentum and spin magnetic moment :-

Since electron possesses an intrinsic spin angular momentum; hence it will also have a spin magnetic moment. Following eqn (10) the spin magnetic moment can be written as

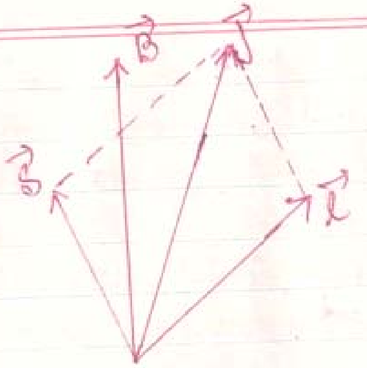
$$\mu_s = -g_s \mu_B S^* \text{ --- } (11)$$

Experimental evidences show that $g_s = 2$ (Explanation is not known) [Stern-Gerlach Expt].

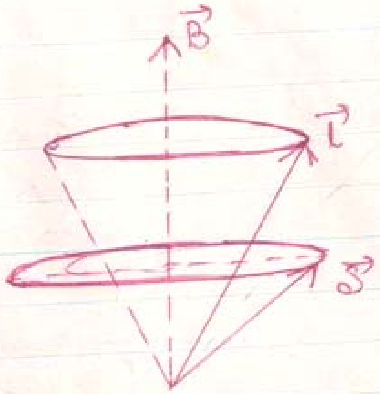
$$\therefore \boxed{\mu_s = -\frac{2e\hbar}{2m}\sqrt{S(S+1)}} \text{ --- } (12)$$



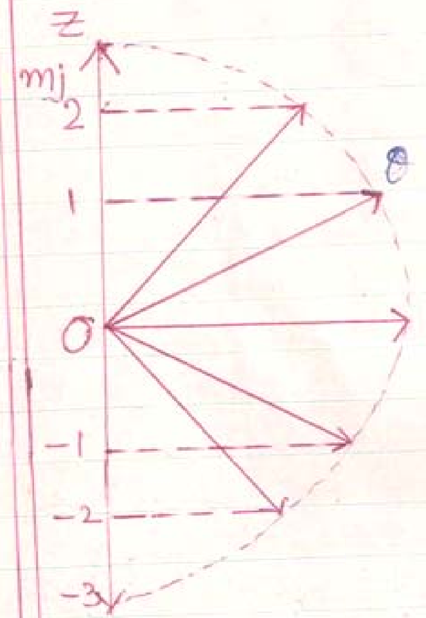
Spin Orbital Interactions :- From the figure we find that $\vec{l}$ and $\vec{S}$ can not be parallel and anti parallel to each other. For a weak external field $\vec{B}$, the vectors precess around $\vec{B}$ and spatially quantised. However as $\vec{B}$ increases the $\vec{l}$ and $\vec{S}$ are uncoupled and precess independently around $\vec{B}$ and are spatially quantised independently. In this case no. $\vec{J}$ is formed and $m_j = m_l + m_s$.



weak field.



Strong field.



$$\boxed{m_j = m_l + m_s}$$

No. $\vec{J}$ vector formed.

$n\,{}^{r}l_j$ $\quad n$ = Total Or Principal Q.No.

for a given $n \rightarrow l = 0, 1, 2 \text{ - - - - } (n-1)$.

l = Orbital Q.No. $\quad 0 = s,\ 1 = p,\ 2 = d$ - - -

$j = l+s$ or $l-s$ $\quad r = 2s+1$ = multiplicity Of the term

Write the term level of Na:

$Na_{11} \rightarrow 1s^2 2s^2 2p^6 \boxed{3s^1}$

$n=3$, $l=0$ Since in the valance orbit we have only one electron $S = 1/2$ (Spin Q.N

Multiplicity of the term $= 2 \times S + 1 = 2$

$j = 0 + \frac{1}{2} = 0 + \frac{1}{2} = 1/2$ $\quad j = l - 1/2 = -1/2$ (

$Na_{11} \rightarrow 3\,{}^2S_{1/2}$

For a single valance electron atom; spin $S = 1/2$ always and multiplicity 'r' is always

Double valence electron atom (in L-S Coupl

$S = S_1 + S_2$ or $S_1 - S_2$

$S = 1/2 + 1/2 = 1$ or $S = 0$

when total spin Quantum numbers 'S' is zero multiplicity $R = 1$.

when total spin Quantum number 'S' is one

$R = 2 \times 1 + 1 = 3$

For d - orbital $l = 2$ $\quad j = l + 1/2 = 5/2$

$j = l - 1/2 = 3/2$

$3\,{}^2D_{5/2}$; $3\,{}^2D_{3/2}$